

# Fumiyuki Kato

Software Engineer at Preferred Networks, Inc.  
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DBLP: [link](#)



## WORK

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**AI/Software Engineer** at Preferred Networks, Inc.

*April 2024 - present*

I work in the Retail Business Division, developing AI-driven operational systems for supermarket operations. I am responsible for end-to-end product development and operations, including optimization design, training and deployment of ML models, design/build/operation of data platforms, and application development and operation (Web and Android). In addition, I regularly support pre-sales activities, manage cross-company collaboration projects, and contribute to commissioned consulting and R&D engagements separate from our core products.

**Visiting Scholar** at Emory University (U.S.)

*April 2023 - September 2023*

Research on privacy-preserving federated learning (collaborated with Prof. Li Xiong [link]).

**JSPS Student Research Fellow**

*April 2023 - March 2024*

Academic research on privacy-enhancing technologies.

## EDUCATION

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**Ph.D. in Informatics** Kyoto University

*April 2021 - March 2024*

Department of Social Informatics, Graduate School of Informatics.

Doctoral Thesis: A Study on Private and Secure Federated Learning. [link]

Supervised by Prof. Yang Cao and Prof. Masatoshi Yoshikawa.

During my three-year Ph.D. program, I published several papers on statistical privacy—particularly differential privacy—including work accepted to VLDB for three consecutive years, a leading conference in database systems.

**Master of Informatics** Kyoto University

*April 2019 - March 2021*

Department of Social Informatics, Graduate School of Informatics.

Master's Thesis: PCT-TEE: Trajectory-based Private Contact Tracing System with Trusted Execution Environment.

(supervised by Prof. Yang Cao and Prof. Masatoshi Yoshikawa)

**Bachelor of Engineering** Kyoto University

*April 2015 - March 2019*

Computer Science, Informatics and Mathematical Science, Faculty of Engineering.  
Bachelor's Thesis: Future Prediction based on Web News Articles with Publication  
Source and Time.  
(supervised by Prof. Makoto P. Kato and Prof. Masatoshi Yoshikawa)

## PUBLICATIONS

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- [1]. **Fumiyuki Kato**, Li Xiong, Shun Takagi, Yang Cao, Masatoshi Yoshikawa. ULDP-FL: Federated Learning with Across Silo User-Level Differential Privacy. In International Conference on Very Large Databases (VLDB), 2024. [paper] [code]
- [2]. Shun Takagi, Li Xiong, **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa. HRNet: Differentially Private Hierarchical and Multi-Resolution Network for Human Mobility Data Synthesis. In International Conference on Very Large Databases (VLDB), 2024. [paper] [code]
- [3]. **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa. OLIVE: oblivious federated learning on trusted execution environment against the risk of sparsification. In International Conference on Very Large Databases (VLDB), 2023. [paper] [code]
- [4]. Shun Takagi, **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa. From Bounded to Unbounded: Privacy Amplification via Shuffling with Dummies. In IEEE Computer Security Foundations Symposium (CSF), 2023. [paper]
- [5]. **Fumiyuki Kato**, Tsubasa Takahashi, Shun Takagi, Yang Cao, Seng Pei Liew, Masatoshi Yoshikawa. HDPView: Differentially Private Materialized View for Exploring High Dimensional Relational Data. In International Conference on Very Large Databases (VLDB), 2022. [paper] [code]
- [6]. Seng Pei Liew, Tsubasa Takahashi, Shun Takagi, **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa. Network Shuffling: Privacy Amplification via Random Walks. In ACM SIGMOD Conference on Management of Data (SIGMOD), 2022. [paper]
- [7]. Shun Takagi, **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa. Asymmetric Differential Privacy. In IEEE International Conference on Big Data (BigData), 2022. [paper]
- [8]. Ruixuan Cao, **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa. An Accurate, Flexible and Private Trajectory-based Contact Tracing System. In Information Integration and Web Intelligence (iiWAS), 2022. [paper]
- [9]. **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa. PCT-TEE: Trajectory-based Private Contact Tracing System with Trusted Execution Environment. ACM Transactions on Spatial Algorithms and Systems (TSAS), 2021. [paper] [code]
- [10]. **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa. Preventing Manipulation Attack in Local Differential Privacy using Verifiable Randomization Mechanism. In 35th Annual IFIP WG 11.3 Conference, DBSec, 2021. [paper] [code]
- [11]. **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa. Secure and Efficient Trajectory-Based Contact Tracing using Trusted Hardware. In PSBD@IEEE BigData, 2020. [paper]
- [12]. Zheng Wen, Keping Yu, Xin Qi, Toshio Sato, Yutaka Katsuyama, Takuro Sato, Wataru Kameyama, **Fumiyuki Kato**, Yang Cao, Masatoshi Yoshikawa, Min Luo, Jun Hashimoto. Blockchain-empowered Contact Tracing for COVID-19 Using Crypto-spatiotemporal Information. In IEEE HEALTHCOM, 2020. [paper]

## SKILLS

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### Programming

*Advanced* Python, AWS  
*Basic* (used in work/research) JavaScript/TypeScript, PyTorch, C++, Rust, Perl, Kotlin, Java, GCP

### Languages

*Native* Japanese  
*Basic* English

## ADDITIONAL EXPERIENCE

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**(Intern) Data Engineer** at Recruit Co.,Ltd. *October 2022 - November 2022*

Implementing and validating dynamic A/B testing.

**(Intern) Software Engineer** at Preferred Networks *August 2022 - September 2022*

Over-the-air (OTA) system.

**(Intern + part-time) Researcher** at LINE Corporation *August 2020 - March 2021*

Research on privacy-enhancing technologies. [blog post]

**(Part-time) Web Application Engineer** at Hatena Inc. *May 2019 - August 2020*

Developing backend and frontend of large-scale web applications mainly using Perl and JavaScript.

**(Intern) Software Engineer** at Recruit Co.,Ltd. *October 2019 - November 2019*

Developing backend and frontend for large-scale web applications mainly using Spring Boot and Java.

**(Intern) Data Scientist** at Wantedly Inc. *September 2019*

Proof of concept for a recommendation feature using Python, Ruby, and BigQuery.

**(Intern) ML Engineer** at M3 Inc. *August 2019*

Developing backend infrastructure for a recommendation service mainly using Scala and Kubernetes.

**(Part-time) Software Engineer** at MediaMaxJapan Inc. *July 2018 - March 2019*

Developing web applications mainly using Kotlin and TypeScript; experienced development strictly following DDD.

**(Intern) Software Engineer** at Kiizan Kiizan Inc. *February 2018 - May 2018*

Developing web applications mainly using Ruby.

## AWARDS

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- (March 2024) The 20th DBSJ Kanbayashi Young Researcher Award.

- (Feb 2022) Excellent Research Award at Kyoto University's 16th ICT Innovation.
- (Mar 2021) DEIM 2021 Student Presentation Award.
- (Feb 2021) Master Thesis Award of the Department of Social Informatics, Kyoto University.
- (Mar 2020) DEIM 2020 Online Presentation Award.

## FUNDING

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- (April 2023 - March 2024) JSPS Student Research Fellow (DC2) (2,400,000 + 1,000,000 JPY / year).
- (December 2022) JSPS Overseas Challenge Program for Young Researchers (1,400,000 JPY).
- (April 2021 - March 2023) Kyoto University AI/Data Science Fellowship (2,100,000 JPY / year).
- (March 2021) Exemption from repayment of scholarships of JASSO due to outstanding achievements.